

DATA FLOW DIAGRAMS (DFD) SPECIFICATION

SMARTSHOP — Digital Shop Management, POS & Khata Ledger System

1. DFD NOTATION STANDARD (Yourdon & DeMarco / Gane & Sarson)

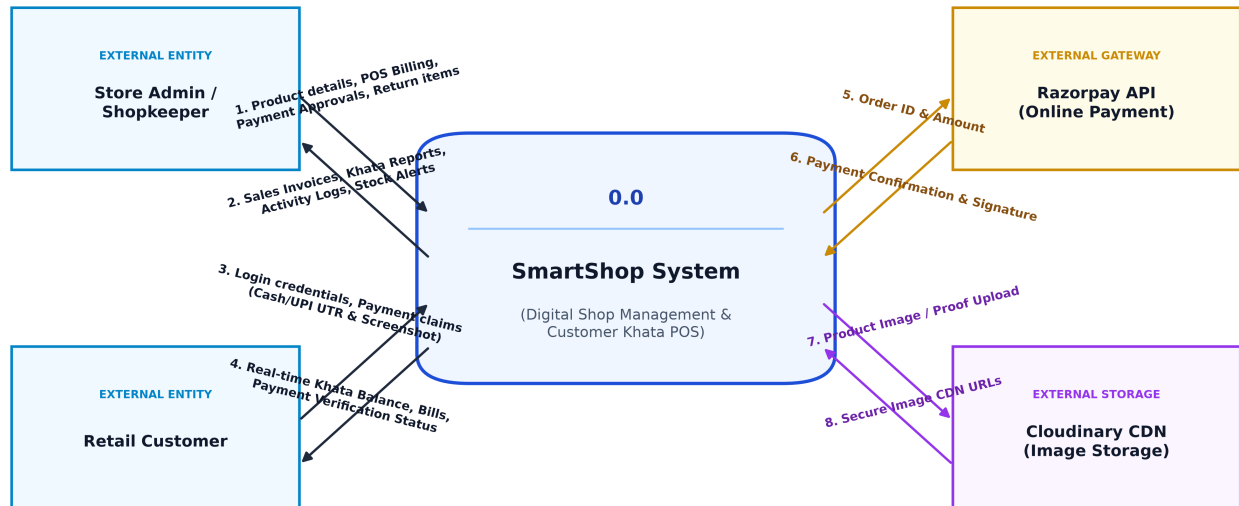
A Data Flow Diagram (DFD) graphically illustrates how information flows through the system, how processes transform inputs into outputs, and where data is persistently stored in the database.

Notation Symbol	DFD Element	SmartShop System Mapping
Square / Rectangle	External Entity	Actors outside software boundary: Store Admin, Retail Customer, Razorpay API, Cloudinary CDN
Rounded Rectangle	Process / Function	Core system logic: 0.0 (SmartShop), 1.0 (Auth), 2.0 (Stock), 3.0 (POS), 4.0 (Khata), 5.0 (Payments), 6.0 (Returns)
Open Parallel Lines	Data Store	MongoDB Collections: D1 (Users), D2 (Customers), D3 (Products), D4 (Sales), D5 (Credits), D6 (Payments), D7 (Returns)
Directed Arrow	Data Flow	Input/Output data paths (credentials, invoices, ledger dues, payment proofs, refund status)

2. DFD LEVEL 0 — CONTEXT LEVEL DIAGRAM

The Level 0 context diagram defines the boundary of the SmartShop system, showing all external actors and their interactions with the central system process.

SmartShop — DFD Level 0 (Context Level Diagram)



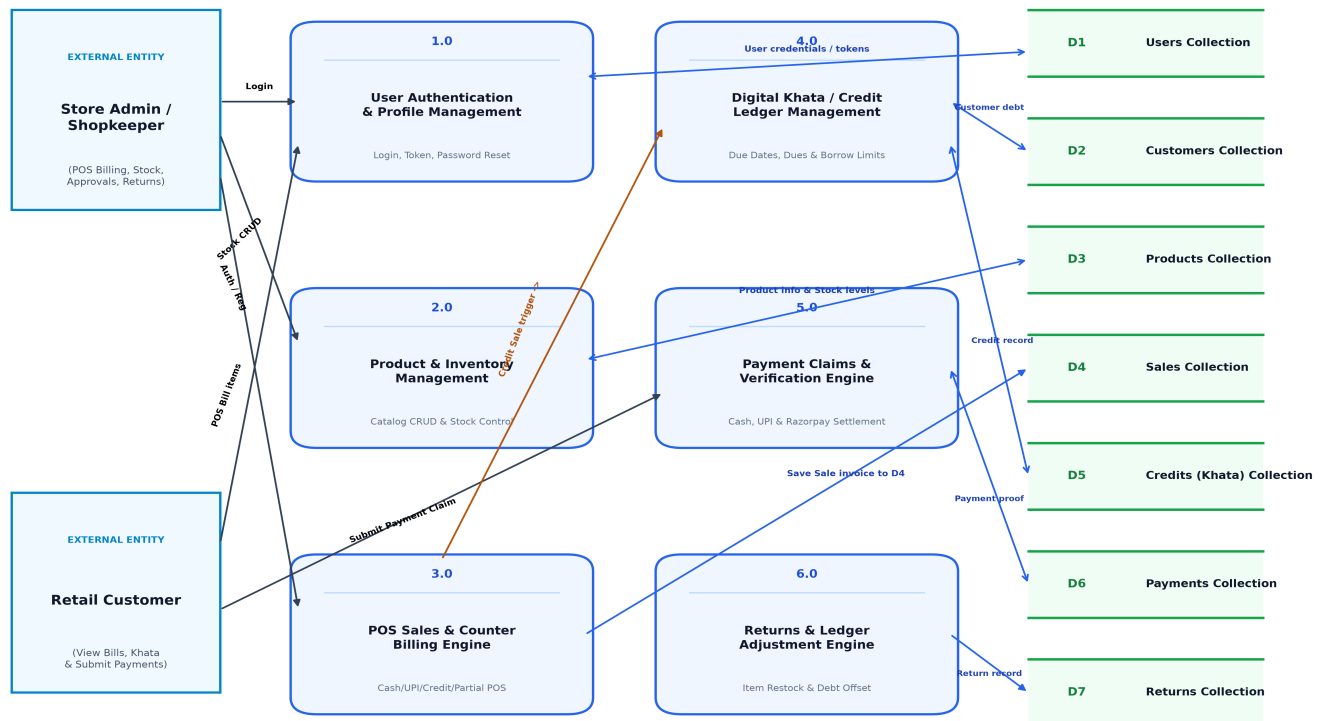
Detailed Data Inflows & Outflows:

- **Store Admin -> System:** Product catalog entries, counter sales items, payment approval/rejection actions, return orders, customer borrow limit updates.
- **System -> Store Admin:** Real-time sales receipts, customer khata ledger reports, activity logs, low stock indicators.
- **Customer -> System:** Login credentials, cash claim submissions, UPI transaction IDs and receipt screenshot proofs, online payment tokens.
- **System -> Customer:** Itemized purchase invoices, real-time pending khata balance, payment clearance status.
- **System <-> Razorpay & Cloudinary:** Razorpay checkout order verification; Cloudinary CDN image streaming & secure URLs.

3. DFD LEVEL 1 — DETAILED FUNCTIONAL DECOMPOSITION

The Level 1 diagram decomposes the system into **6 core processes** and maps their direct interactions with the **7 MongoDB collections (D1 to D7)**.

SmartShop — DFD Level 1 (Detailed Functional Decomposition)



Process Explanations & Database Interactions:

- **1.0 User Authentication & Profile:** Handles login and role-based access. Reads/writes **D1 (Users)** and links customer profiles in **D2 (Customers)**.
- **2.0 Product & Inventory Management:** Manages product catalog, prices, units, and Cloudinary CDN URLs in **D3 (Products)**.
- **3.0 POS Sales & Counter Billing:** Builds counter bills, performs atomic stock deduction in **D3 (Products)**, saves sales in **D4 (Sales)**, and triggers credit creation in Process 4.0 if sale is on credit.
- **4.0 Digital Khata / Credit Ledger:** Tracks borrowed amounts, repayment deadlines, borrow limit validation, and updates debt in **D2 (Customers)** and **D5 (Credits)**.
- **5.0 Payment Claims & Verification:** Saves payment claims in **D6 (Payments)** and decrements customer pending dues atomically in **D5 (Credits)** and **D2 (Customers)** upon admin approval or Razorpay confirmation.
- **6.0 Returns & Ledger Adjustment:** Restores item quantities in **D3 (Products)**, saves return records in **D7 (Returns)**, and deducts outstanding credit debt in **D5 (Credits)**.